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The emergence of the 'planetary boundaries' concept in international environmental law: A proposal for a framework convention

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Grounded in the application of the precautionary principle, the planetary boundaries framework attempts to define a safe operating space where human activities can take place without the risk of transgressing the Earth system's ecological thresholds, thus maintaining the planet's Holocene-like conditions. Since its emergence in 2009, the scientific concept of planetary boundaries has gradually gained recognition in the political arena. This concept has been introduced in the European Union's legal instruments, and it has been included in documents issued at the request of or by the United Nations Secretary-General. However, the concept has not yet been integrated into the legal documents of the United Nations' organs. This article argues for the adoption of a framework convention on planetary boundaries as an effective means for integrating the planetary boundaries approach into international law.

1 | THE EMERGENCE OF THE SCIENTIFIC CONCEPT OF PLANETARY BOUNDARIES

Building on the extraordinary progress of Earth system science during the last few decades, in 2009, a group of 29 scientists led by Johan Rockström attempted to define the borders of what they called 'the safe operating space for humanity', meaning the space in which human activities can take place without the risk of transgressing the Earth system's ecological thresholds.[1] The term 'planetary boundaries' designates these borders. The purpose of this proposal was not to protect the planet as such but rather to protect the state of the planet that has been favourable for the life and development of contemporary human societies, that is, the state of the planet during the Holocene, the geological epoch of the last 11,700 years,[2] which has been characterized by its climatic stability.[3] The group of scientists argued that by organizing to stay within these boundaries, humanity could avoid weakening the Earth system's resilience,[4] which is subject to significant man-made pressures.[5]

Rockström's team identified the following nine biophysical processes of the Earth system that determine the planet's ability to auto-regulate and therefore to maintain its Holocene-like conditions:

[1] J Rockström et al, 'Planetary Boundaries: Exploring the Safe Operating Space for Humanity' (2009) 14 Ecology and Society 32.

[2] R Monastersky, 'The Human Age' (2015) 519 Nature 145.

[3] W Dansgaard et al, 'Evidence for General Instability of Past Climate from a 250-kyr Ice-core Record' (1993) 364 Nature 218, 220.

[4] Resilience has been defined in various manners by different scientific disciplines. See FS Brand and K Jax, 'Focusing the Meaning(s) of Resilience: Resilience as a Descriptive Concept and a Boundary Object' (2007) 12 Ecology and Society 23. The following are two examples of definitions that have been applied to ecological systems: 'In ecology and biology, resilience is the capacity of an ecosystem, of a species or of an individual to recuperate a normal functioning or development after having experienced a disturbance' (free translation of S Bosi and A Euzen (eds), *Prospective Droit, Écologie et Économie de la Biodiversité: Une Prospective du CNRS* (CNRS 2013) 62); 'Resilience is the capacity of a system to absorb disturbance and reorganize while undergoing change so as to still retain essentially the same function, structure, identity, and feedbacks' (B Walker et al, 'Resilience, Adaptability and Transformability in Social-ecological Systems' (2004) 9 Ecology and Society 5).

[5] Scientists argue that we have entered a new geological era, 'the Anthropocene', in which humans have become a geophysical force of the planetary system. See W Steffen et al, 'The Anthropocene: From Global Change to Planetary Stewardship' (2011) 40 Ambio 739. According to these authors (ibid 755), 'the Anthropocene is a dynamic state of the Earth System, characterized by global environmental changes already significant enough to distinguish it from the Holocene, but with a momentum that continues to move it away from the Holocene at a geologically rapid rate'. Regarding the need to think differently concerning environmental law in the Anthropocene, see LJ Kotzé, 'Rethinking Global Environmental Law and Governance in the Anthropocene' (2014) 32 Journal of Energy and Natural Resources Law 121.

climate change, stratospheric ozone depletion, land system change, global freshwater use, rate of biodiversity loss, ocean acidification, interference with phosphorus and nitrogen cycles, atmospheric aerosol loading and chemical pollution.

Three of these nine key processes of the Earth system are directly associated with ecological thresholds, or 'tipping points',[6] at the global level: climate change, ocean acidification and stratospheric ozone depletion. The transgression of these processes would provoke abrupt, nonlinear and potentially irreversible environmental changes, driving the planet toward a different state, with consequences for humanity that remain unknown. According to the scientists, although the other six processes are not associated with global ecological thresholds, they may have the capacity, because of their intensity and their cumulative effects, to trigger or favour the breaching of global thresholds in other processes.

For each of the nine biophysical processes identified, the scientists attempted to define a quantified planetary boundary. With this goal in mind, they chose at least one control variable for each process, and for each variable, they determined a ceiling value representing its boundary level. As an example, one of the two control variables chosen for climate change was the concentration of carbon dioxide in the atmosphere, and its ceiling value was fixed at 350 parts per million (ppm). Quantification was possible for all processes except atmospheric aerosol loading and chemical pollution.[7]

The ceiling values were determined by taking into consideration what the scientists estimated to be a safe distance with regard to the global ecological threshold for the processes associated with such a threshold or with regard to a certain dangerous level for the other processes. In the first case, a zone of uncertainty exists regarding the exact position of the ecological threshold. In the second case, the zone of uncertainty is associated with the dangerous level. Consequently, by applying the precautionary principle, which constitutes one of the pillars of the planetary boundaries approach, each value was chosen to place the planetary boundary at the lowest level of the scientific uncertainty zone.

Thus, global ecological thresholds and planetary boundaries are used as different concepts, and their difference must be clear in order to understand the planetary boundaries framework. One of the main differences that should be kept in mind is that while global ecological thresholds are biophysical phenomena whose exact positions are determined by biophysical processes, planetary boundaries are a human creation and their positions are therefore determined by humans.[8]

The planetary boundaries framework highlights the interdependence between the various key processes of the Earth system that have been identified.[9] Since each planetary boundary level has been determined by assuming that the other boundaries are not exceeded, it has been stressed that exceeding one boundary could lead to either a shift in the position or a transgression of other boundaries.

In 2015, a new article was published in which the original planetary boundaries were revised and updated.[10] The study suggests that among the seven key global biophysical processes for which reference values could be identified, four have already entered the uncertainty zone: climate change, land system change, biosphere integrity (formerly 'rate of biodiversity loss' in the 2009 version and now including genetic and functional diversity) and biogeochemical flows ('interference with phosphorus and nitrogen cycles' in the previous version). It is possible that the latter two processes have already entered a high-risk zone. Furthermore, the authors of this updated version insist that interactions occur between various processes at all scales (local, regional and global).

The aim of the planetary boundaries approach (i.e., to maintain the planet's Holocene-like conditions, the only conditions that we know for certain are favourable for the life and well-being of contemporary human societies) is consistent with the idea defended by fields of study such as ecological economics, according to which nature is the substratum on which society and the economy lie.[11] For this reason, we argue that the planetary boundaries framework revisits the classic paradigm of sustainable development issued by the Brundtland Report,[12] which aimed to establish a balance between environmental, social and economic interests placed on an equal footing. In this sense, some authors assert that the planetary boundaries concept 'suggests a hierarchical order for the elements of sustainable development: the biophysical environment comes first, and human society and the economy second'.[13]

Nevertheless, the planetary boundaries framework has also been criticized within the scientific community. Some critics address the negative implications that such an approach may have in terms of

[6]The terms 'ecological thresholds' and 'tipping points' are often used interchangeably and can be defined in multiple manners. We will only recall here one of the existing definitions of a 'tipping point': 'In ecology, a critical point beyond which the system passes abruptly from one state to another. In the theory of dynamic systems, the boundary of the basin of attraction or critical bifurcation value of a parameter' (free translation of Bosi and Euzen (n 4) 63). See also JC Rocha, R Biggs and GD Peterson, 'Regime Shifts: What Are They and Why Do They Matter?' (2014) <http://www.regimeshifts.org/datasets-resources/details/15/26>.

[7]See, e.g., the difficulties found with respect to chemical pollution: L Diamond et al, 'Exploring the Planetary Boundary for Chemical Pollution' (2015) 78 Environment International 8.

[8]Frank Biermann underlines that the definition of planetary boundaries is, ultimately, a political process, a social construct resulting from negotiations in the context of a very differentiated global society. F Biermann, 'Planetary Boundaries and Earth System Governance: Exploring the Links' (2012) 81 Ecological Economics 4, 6.

[9]In this regard, see also M Nilsson and Å Persson, 'Can Earth System Interactions be Governed? Governance Functions for Linking Climate Change Mitigation with Land Use, Freshwater and Biodiversity Protection' (2012) 75 Ecological Economics 61.

[10]W Steffen et al, 'Planetary Boundaries: Guiding Human Development on a Changing Planet' (2015) 347 Science 1259855.

[11]More precisely, society lies within nature, and the economy lies within society. See R Costanza et al, *Building a Sustainable and Desirable Economy-in-Society-in-Nature* (United Nations Division for Sustainable Development 2012).

[12]'Report of the World Commission on Environment and Development: Our Common Future' UN Doc A/42/427 (4 August 1987).

[13]RE Kim and K Bosselmann, 'International Environmental Law in the Anthropocene: Towards a Purposive System of Multilateral Environmental Agreements' (2013) 2 Transnational Environmental Law 285.

power relations, justice and respect for human rights.[14] In particular, Kate Raworth underlines the need to combine this approach with concerns regarding human rights and social justice.[15] Furthermore, some authors have underlined the pernicious effects of using thresholds in environmental topics,[16] whereas others have noted the need to adopt a multi-scale approach.[17] Finally, the choice of biophysical processes and the fixed values for planetary boundaries, in addition to the manner in which these values were fixed, have been subject to discussion.[18]

2 | INTRODUCTION OF THE PLANETARY BOUNDARIES CONCEPT IN THE INTERNATIONAL POLITICAL ARENA

After appearing in and spreading among the scientific community, the planetary boundaries concept gradually began to be introduced in official (legal and non-legal) documents.

First, the concept was included in documents issued at the request of or by the United Nations Secretary-General. The concept was mentioned for the first time in the Report of the High-level Panel of the Secretary-General on Global Sustainability, entitled 'Resilient People, Resilient Planet: A Future Worth Choosing', which was submitted to the Secretary-General on 30 January 2012.[19] This report contains several references to planetary boundaries[20] that mainly address the panel's long-term vision,[21] along with recommendation 51. The latter invites governments and the scientific community to

> *take practical steps, including the launching of a major global scientific initiative, to strengthen the interface between policy and science. This should include the preparation of regular assessments and digests of the science around such concepts as 'planetary boundaries', 'tipping points' and 'environmental thresholds' in the context of sustainable development.*[22]

It could be tempting to suggest that the essence of the planetary boundaries concept was already present in the 1987 Brundtland Report, which seems to have been the possible intention of the High-level Panel.[23] The Brundtland Report mentions the existence of 'thresholds that cannot be crossed without endangering the basic integrity of the system',[24] and after warning that 'today we are close to many of these thresholds', the report affirms that 'we must be ever mindful of the risk of endangering the survival of life on Earth'.[25] The report also mentions threats such as 'climate change, ozone depletion and species loss' as examples of cases in which 'we may already be close to transgressing critical thresholds' at a global scale.[26] However, in contrast to the planetary boundaries framework, the proposals made in the Brundtland Report for achieving sustainable development were grounded neither in a vision of natural resource management based on ecological systems' *resilience* (the word 'resilience' appeared only once in the report and was used to indicate the resilience of industrial world cities)[27] nor in a conception of ecosystems as dynamic and complex systems. By contrast, the recommendations of the Brundtland Report were grounded in a conventional approach to natural resource management based on ideas of optimization and efficiency, in which nature is in balance and humans can completely control it and predict changes. Currently, the scientific community agrees on the fact that nature is not in balance but rather constantly evolving;[28] nature is not static but rather dynamic.[29]

Despite the call made by the Secretary-General encouraging Member States 'to consider the recommendations thoroughly and to

[14]See, e.g., M Leach, 'Resilience 2014: Limits Revisited? Planetary Boundaries, Justice and Power' (2014) <http://steps-centre.org/2014/blog/resilience2014-leach/>.

[15]K Raworth, 'A Safe and Just Space for Humanity: Can We Live within the Doughnut?' (Oxfam 2012) <https://www.oxfam.org/sites/www.oxfam.org/files/dp-a-safe-and-just-space-for-humanity-130212-en.pdf>. The combination of the concept of biophysical limits of the planet with social equity considerations has been presented by other authors as a condition for global sustainability: W Steffen and M Stafford Smith, 'Planetary Boundaries, Equity and Global Sustainability: Why Wealthy Countries Could Benefit from More Equity' (2013) 5 Current Opinion in Environmental Sustainability 403. To avoid incompatibility with other important goals, such as human development and social justice, Biermann argues that the concept of planetary boundaries must serve as a policy target only within the larger context of sustainable development. Biermann (n 8) 5.

[16]SL Lewis, 'We Must Set Planetary Boundaries Wisely' (2012) 485 Nature 417. WH Schlesinger, 'Planetary Boundaries: Thresholds Risk Prolonged Degradation' (2009) 3 Nature Reports Climate Change 112: 'Unfortunately, policymakers face difficult decisions, and management based on thresholds, although attractive in its simplicity, allows pernicious, slow and diffuse degradation to persist nearly indefinitely.'

[17]Regarding the need to take a multi-scale approach, see Nilsson and Persson (n 9). This criticism was taken into account in the 2015 version of planetary boundaries. See, in response, TP Hughes et al, 'Multiscale Regime Shifts and Planetary Boundaries' (2013) 28 Trends in Ecology and Evolution 389. Concerning the efforts to implement the framework at the local level, see U Pisano and G Berger, 'Planetary Boundaries for SD, From an International Perspective to National Applications' (2013) 30 ESDN Quarterly Report (this paper gathers the official points of view of eight European countries regarding the planetary boundaries concept and gives concrete examples of its implementation in Sweden); T Hayha et al, 'From Planetary Boundaries to National Fair Shares of the Global Safe Operating Space: How Can the Scales be Bridged?' (2016) 40 Global Environmental Change 60.

[18]Regarding the rate of biodiversity loss, see G Mace et al, 'Approaches to Defining a Planetary Boundary for Biodiversity' (2014) 28 Global Environmental Change 289.

[19]'Report of the High-level Panel of the Secretary-General on Global Sustainability: Resilient People, Resilient Planet: A Future Worth Choosing' UN Doc A/66/700 (1 March 2012). See also the published version: United Nations Secretary-General's High-level Panel on Global Sustainability, *Resilient People, Resilient Planet: A Future Worth Choosing* (United Nations 2012) <http://en.unesco.org/system/files/GSP_Report_web_final.pdf>.

[20]See, e.g., 'Report of the High-level Panel of the Secretary-General on Global Sustainability' (n 19) paras 6, 17, 31, 251 and 255.

[21]ibid para 6: 'The long-term vision of the High-level Panel on Global Sustainability is to eradicate poverty, reduce inequality and make growth inclusive, and production and consumption more sustainable, while combating climate change and respecting a range of other planetary boundaries.'

[22]ibid para 255.

[23]ibid para 31.

[24]'Report of the World Commission on Environment and Development' (n 12) Chapter 1, 'A Threatened Future', para 23.

[25]ibid.

[26]ibid para 32.

[27]ibid Chapter 9, 'The Urban Challenge', para 23.

[28]See, e.g., N Hervé-Fournereau et al, 'Résilience, Irréversibilités et Incertitudes' in Bosi and Euzen (n 4) 25; CS Holling, 'Engineering Resilience Versus Ecological Resilience' in P Schulze (ed), *Engineering within Ecological Constraints* (National Academy Press 1996) 31.

[29]C. Folke et al, 'Resilience Thinking: Integrating Resilience, Adaptability and Transformability' (2010) 15 Ecology and Society 20.

inscribe those that are ripe for immediate decision in the outcome document of the United Nations Conference on Sustainable Development (Rio+20), which is currently being negotiated',[30] no reference to the concept of planetary boundaries was made in the aforementioned outcome document. The concept, which was missing in the original zero-draft version,[31] was included in later versions of the negotiation text[32] and faced opposition by some countries, especially developing countries,[33] the United States and Canada.[34] A reference to planetary boundaries remained in the negotiation document of 2 June 2012.[35] Nevertheless, and despite support expressed by civil society groups,[36] the concept was not included in the outcome document of the conference, entitled 'The Future We Want'.[37]

The second time that a reference to the planetary boundaries concept was included in documents issued at the request of or by the United Nations Secretary-General was in his report to the United Nations General Assembly on agriculture development, food security and nutrition of 13 August 2013.[38] Notably, not only were some of the main conclusions of Rockström's team acknowledged but also, for the first time, there was an attempt to define the concept. The report refers to planetary boundaries as boundaries 'within which humanity can exist without causing irreparable environmental damage'.[39]

The United Nations Secretary-General referred once more to the planetary boundaries concept in the synthesis report on the post-2015 agenda, entitled 'The Road to Dignity by 2030: Ending Poverty, Transforming All Lives and Protecting the Planet', presented to the United Nations General Assembly on 4 December 2014.[40] In this document, planetary boundaries are considered part of one of the six essential elements for delivering on the sustainable development goals, with the essential element 'Planet: to protect our ecosystems for all societies and our children.' More specifically, regarding this element, the report lists a series of actions that are necessary 'to respect our planetary boundaries'.[41]

Notwithstanding the willingness of the Secretary-General to promote the planetary boundaries concept in the international political arena, the concept has not yet been integrated into the legal documents (binding or not) of the United Nations organs. In this regard, the 2030 Agenda for Sustainable Development, entitled 'Transforming Our World', which establishes the sustainable development goals and was adopted by the General Assembly on 25 September 2015,[42] does not make explicit reference to the concept.

The 2030 Agenda clearly places the 'planet' at the same rank as 'people',[43] from which we can undoubtedly affirm that the protection of the planet is treated as an issue of major international concern. However, the Agenda does not specify that what is at stake is the maintenance of the Earth system in a Holocene-like state, which

[30]'Report of the High-level Panel of the Secretary-General on Global Sustainability' (n 19) para 7.

[31]'Zero Draft of the Outcome Document of the United Nations Conference on Sustainable Development (UNCSD)' (10 January 2012) <http://www.rio20.gov.br/documents/esboco-zero-do-documento-final-da-conferencia/at_download/zero-draft.pdf>.

[32]See, e.g., the zero-draft version of 17 April 2012, 11:00 am: 'Negotiated Draft as of 28 March 2012 with Co-Chairs Suggested Text' (17 April 2012) <https://fr.scribd.com/document/98253647/116Negotiation-Text-w-CoChairs-Suggested-Text-Complete-170412-1100>: Preamble, '[1. ter […] We join forces to eradicate poverty, reduce inequality, make growth inclusive, create new jobs and promote sustainable production and consumption, while combating climate change and respecting other planetary boundaries.… – EU]' (ibid 3–4).

In para 4 of the preamble, the European Union proposed to replace 'while preserving and protecting the life support system of our common home, our shared planet' with 'whilst integrating the need to protect, preserve and restore the natural resources of our shared planet, within the carrying capacity of its ecosystems and planetary boundaries' (ibid 7).

Section III 'Green Economy in the context of sustainable development and poverty eradication', '[Pre25.non We are convinced that the green economy is a means to achieving our common vision to eradicate poverty, reduce inequality and make growth inclusive, and [production and consumption more sustainable, while combating climate change and respecting a range of other planetary boundaries/promote protection of the environment and natural resources – US].…]' (ibid 42–43).

Section III 'Green Economy in the context of sustainable development and poverty eradication', '[26 bis We recognize that to achieve sustainable growth in a way that respects resource constraints and [planetary boundaries – Canada] …]' (ibid 49).

[33]A commentator on the Rio+20 process stated that the reason why the concept of planetary boundaries was discarded from the negotiation text was 'partly due to concerns from some poorer countries that its adoption could lead to the sidelining of poverty reduction and economic development' and partly 'because the idea is simply too new to be officially adopted, and needed to be challenged, weathered, and chewed over to test its robustness before standing a chance of being internationally accepted at UN negotiations'. A Irwin, 'Your Guide to Science and Technology at Rio+20' (SciDev.Net, 12 June 2012) <http://www.scidev.net/global/climate-change/feature/your-guide-to-science-and-technology-at-rio-20-1.html>. During the UNCSD Third Intersessional Meeting, on 26 March 2012, the G77 and China group asked for clarification of the text on 'planetary boundaries'. D Paul et al, 'UNCSD Third Intersessional Meeting: Monday, 26 March 2012' (2012) 27 Earth Negotiations Bulletin 7.

[34]During the UNCSD 'informal-informal' consultations, on 19 March 2012, both the United States and Canada requested to delete references to 'planetary boundaries'. D Paul et al, 'UNCSD Informal Consultations: Monday, 19 March 2012' (2012) 27 Earth Negotiations Bulletin 2.

[35]UNCSD negotiation document (2 June 2012) <https://fr.scribd.com/document/96419644/Draft-of-UN-Rio-20-main-text> para 42: 'We recognize the important contribution of the scientific and technological community to sustainable development. We are committed to working with and fostering collaboration among academic, scientific and technological community, in particular in developing countries, to close the technological gap between developing and developed countries, strengthen the science–policy interface as well as to foster international research collaboration [including in the area of planetary boundaries – EU; US, G77 delete].'

[36]See P Doran et al, UN Conference on Sustainable Development: Wednesday, 20 June 2012' (2012) 27 Earth Negotiations Bulletin 9.

[37]Outcome of the UNCSD: 'The Future We Want' UN Doc A/CONF.216/L.1 (19 June 2012) <https://rio20.un.org/sites/rio20.un.org/files/a-conf.216l-1_english.pdf.pdf>.

[38]'Report of the Secretary-General on Agriculture Development, Food Security and Nutrition' UN Doc A/68/311 (13 August 2013).

[39]ibid para 12: 'Since around 1950, the acceleration of industrialization and human development has imposed steadily increasing pressure on the biophysical Earth systems. Depletion of finite natural resources, combined with a decreasing capacity of ecosystems to absorb the waste of human activity, has prompted scientists and policymakers to identify and quantify "planetary boundaries" within which humanity can exist without causing irreparable environmental damage. Planetary boundaries have already been exceeded in the areas of climate change, nitrogen and phosphorus cycles and biodiversity loss; in particular, climate projections based on existing data forecast a global mean surface warming of more than 2°C becoming the norm by 2060.'

[40]'Synthesis Report of the Secretary-General on the Post-2015 Agenda: The Road to Dignity by 2030: Ending Poverty, Transforming All Lives and Protecting the Planet' UN Doc A/69/700 (4 December 2014).

[41]ibid para 75.

[42]UNGA 'Transforming Our World: The 2030 Agenda for Sustainable Development' UN Doc A/RES/70/1 (21 October 2015).

[43]ibid preamble: 'This Agenda is a plan of action for people, planet and prosperity'; 'the Goals and targets will stimulate action over the next 15 years in areas of critical importance for humanity and the planet'. ibid para 51: 'What we are announcing today – an Agenda for global action for the next 15 years – is a charter for people and planet in the twenty-first century.'

is the overarching purpose of the planetary boundaries framework. Furthermore, the Agenda does not make any reference to the scientific concern that this system may tip over abruptly and irreversibly toward another state, which would be unfavourable for humanity, if the ecological thresholds are transgressed as a result of human-caused transformations. These omissions indicate the absence of important elements of the planetary boundaries framework in the Agenda. This absence is even more obvious in the Sustainable Development Goals, which do not make any reference at all to the planet. However, the inclusion of the planetary boundaries concept in the Agenda, particularly in its Sustainable Development Goals, had been suggested during the process of shaping a post-2015 agenda.[44]

At the European level, the concept began its incursion into legal instruments in 2012, when recital E of the European Parliament resolution of 24 May 2012 'on a resource-efficient Europe'[45] made reference to 'switching the economy on to a resource-efficient path which respects planetary boundaries'. Subsequently, the Parliament's resolution of 12 March 2013 'on improving the delivery of benefits from EU environment measures: building confidence through better knowledge and responsiveness'[46] recalled in paragraph 28 that 'a common effort is vital to ensure that the EU economy grows in a way that respects resource constraints and planetary boundaries'. Paragraph 11 of the Parliament's resolution of 13 June 2013 'on the Millennium Development Goals – defining the post-2015 framework'[47] also used the expression 'planetary boundaries' in its English,[48] Spanish and Italian versions but, surprisingly, not in the French version. In addition, the European Commission has regularly made reference to 'planetary boundaries' in its working papers and communications.[49]

However, perhaps the most important step forward has been the inclusion of the planetary boundaries concept in the 7th Environment Action Programme (EAP), which was adopted by the decision of the European Parliament and of the Council of 20 November 2013 'on a General Union Environment Action Programme to 2020 "Living Well, within the Limits of Our Planet"'.[50] Even though the 2050 vision expressed in paragraph 1 of the 7th EAP specifically mentioned 'the planet's ecological limits'[51] and, therefore, global thresholds or tipping points, which differ from the planetary boundaries concept as defined by Rockström and colleagues, the expression 'planetary boundaries' was used in other parts of the 7th EAP, and explicit reference was made to the work by Rockström's team. In this regard, one of the 7th EAP's introductory paragraphs[52] and its corresponding footnote[53] made specific reference to some of the main elements grounding the planetary boundaries framework. Furthermore, other references to 'planetary boundaries' were made in the context of the 7th EAP's Priority objective 5 about improving 'the knowledge and evidence base for Union environment policy'[54] and in the context of Priority objective 9 about 'increasing the Union's effectiveness in addressing international environmental and climate-related challenges'.[55]

Notwithstanding the foregoing, no specific content has been assigned to the planetary boundaries concept in the European Union's legal instruments to operationalize the concept. Although the Union has recognized the paramount importance of not transgressing planetary boundaries, it has not taken further steps toward deciding which of the Earth system's biophysical processes determine the ability of the planet to maintain its Holocene-like conditions or toward determining control variables and ceiling values for each of these processes. It seems clear from Priority objective 5 of the 7th EAP that the European Union aims for more progress to be made in terms of scientific knowledge before making any further

[44]See, e.g., the report of the thematic consultation on environmental sustainability in the post-2015 agenda: 'Breaking Down the Silos: Integrating Environmental Sustainability in the Post-2015 Agenda' (2013) <http://www.pnuma.org/sociedad_civil/reunion2013/documentos/POST-2015/2013%20Integrating%20Environmental%20Sustainability%20Post-2015.pdf>; 'Synthesis Report of the Secretary-General on the Post-2015 Agenda' (n 40); and D Griggs et al, 'Sustainable Development Goals for People and Planet' (2013) 495 Nature 305.

[45]Parliament Resolution (EP) P7_TA(2012)0223 of 24 May 2012 on a resource-efficient Europe [2013] OJ C264E/59.

[46]Parliament Resolution (EP) P7_TA(2013)0077 of 12 March 2013 on improving the delivery of benefits from EU environment measures: building confidence through better knowledge and responsiveness [2016] OJ C36/43.

[47]Parliament Resolution (EP) P7_TA(2013)0283 of 13 June 2013 on the Millennium Development Goals – defining the post-2015 framework [2016] OJ C65/136.

[48]ibid para 11: The European Parliament 'urges that poverty eradication, which is the primary objective of EU development cooperation, and the achievement of sustainable social and environmental development within the planetary boundaries must be the imperative global priorities for the post-2015 development agenda'.

[49]For example, Commission (EU), 'Roadmap to a Resource Efficient Europe' (Communication) COM(2011) 571 final, 20 September 2011; Commission (EU), 'Consultative Communication on the Sustainable Use of Phosphorus' (Communication) COM(2013) 517 final, 8 July 2013; Commission (EU), 'A Decent Life for All: From Vision to Collective Action' (Communication) COM(2014) 335/final, 2 June 2014; and Commission staff working document 'EU Assessment of Progress in Implementing the EU Biodiversity Strategy to 2020' SWD (2015) 187 final, 2 October 2015.

[50]Parliament and Council Decision (EC) 1386/2013/EU on a General Union Environment Action Programme to 2020 'Living Well, within the Limits of Our Planet' [2013] OJ L354/171.

[51]ibid para 1: 'In 2050, we live well, within the planet's ecological limits.' See also ibid para 10: 'To live well in the future, urgent, concerted action should be taken now to improve ecological resilience and maximize the benefits environment policy can deliver for the economy and society while respecting the planet's ecological limits.'

[52]ibid para 8: '… there is evidence that planetary boundaries for biodiversity, climate change and the nitrogen cycle have already been transgressed'.

[53]ibid footnote 3: 'Thresholds associated with nine "planetary boundaries" have been identified which, once crossed, could lead to irreversible changes with potentially disastrous consequences for humans, including: climate change, biodiversity loss, global freshwater use, ocean acidification, the nitrogen and phosphorus cycles and land-use change (Ecology and Society, Vol. 14, No 2, 2009).'

[54]For example, in ibid para 71.1, the 7th EAP highlighted the need to pay particular attention to data and knowledge gaps and argued both that 'advanced research is required to fill such gaps and adequate modelling tools are needed to better understand complex issues related to environmental change, such as the impact of climate change and natural disasters, the implications of species loss for ecosystem services, environmental thresholds and ecological tipping points' and that 'further research into planetary boundaries, systemic risks and our society's ability to cope with them will support the development of the most appropriate responses'. Related to this, ibid para 73 indicated that improving the knowledge and evidence base for the Union's environment policy requires '(i) coordinating, sharing and promoting research efforts at Union and Member State level with regard to addressing key environmental knowledge gaps, including the risks of crossing environmental tipping points and planetary boundaries'.

[55]In ibid para 106, the 7th EAP indicated that attaining Priority objective 9 required '(viii) ensuring that economic and social progress is achieved within the carrying capacity of the Earth, by increasing understanding of planetary boundaries, inter alia, in the development of the post-2015 framework in order to secure human well-being and prosperity in the long-term'.

decisions on the subject, despite recognizing in paragraph 10 that urgent action should be taken in order to live well in the future while respecting the planet's ecological limits.

Nevertheless, as mentioned above, the reference to the planetary boundaries concept in the 7th EAP is unquestionable. Therefore, it is convenient to determine the legal value that the concept has according to European law. As we mentioned previously, the 7th EAP was adopted by a decision of the European Parliament and of the Council. This type of decision is legally binding for Member States of the European Union. In this respect, Article 3 of this decision mentions that 'the relevant Union institutions and Member States are responsible for taking appropriate action, with a view to the delivery of the priority objectives set out in the 7th EAP'. However, none of those priority objectives are specifically aimed at the non-transgression of planetary boundaries.

It is unclear, however, what the real priority of the planetary boundaries concept is in a European policy essentially centred on economic growth, even if the latter is qualified as 'sustainable'. In this regard, the communication from the Commission of 3 March 2010 entitled 'Europe 2020: A Strategy for Smart, Sustainable and Inclusive Growth'[56] defines 'sustainable growth' as 'promoting a more resource efficient, greener and more competitive economy'. As was the case for the Brundtland Report, this strategy is grounded in an outdated vision of natural resource management based on ideas of optimization and efficiency and omits concepts such as resilience, dynamism and the complexity of ecosystems. Indeed, the term 'ecosystem' is not mentioned even once in this document.

The foregoing analysis demonstrates that even though the planetary boundaries concept is penetrating the international political arena, it has not yet been included in international legal instruments, except in the European Union framework. One possible explanation for this observation is Biermann's acknowledgement that the actual lack of precision in the scientific basis of some boundaries limits the present use of the concept for political decision-making.[57]

The integration of the concept of planetary boundaries into international environmental law would nevertheless be of major interest. Several arguments plead in favour of such integration. Biophysical processes that determine the ability of the Earth system to self-regulate have been treated on a sectoral basis despite their interdependency, as shown by the variety of existent legal regimes (e.g., climate,[58] biodiversity,[59] ozone layer[60]). It has been demonstrated that decisions or actions aimed at avoiding or reducing the impacts of human activities in a given area can result in unintended negative consequences in another sector.[61] For this reason, incorporation of the planetary boundaries framework in international law could help strengthen the interactions between different legal regimes and intensify the coordination and cooperation between responsible institutions.

However, the concept of a 'safe operating space for humanity', which is part of the planetary boundaries framework, could become a global common goal that brings together all efforts made in the environmental field. As a global common goal, this concept would foster the emergence of new rules in pursuit of this objective and adaptation of the existing law.[62] Judges would therefore be able to rely on the concepts of 'safe operating space for humanity' and 'planetary boundaries' for the interpretation of existing rules. Furthermore, the fact that planetary boundaries consist of human-determined ceiling values for given control variables makes the framework operational and thus attractive to decision-makers.

3 | PROPOSAL FOR A FRAMEWORK CONVENTION ON PLANETARY BOUNDARIES

Despite the particular interest of the scientific community in the planetary boundaries framework, there are different points of view regarding how this framework can be better incorporated into international law.

For example, Biermann argues in favour of effectively governing specific types of social behaviour that may contribute to the violation of a planetary boundary rather than creating one global institution for each boundary.[63] He gives the example of the Convention on International Trade in Endangered Species of Wild Fauna and Flora regarding biodiversity.[64] He also argues that normative and institutional conflicts emerging from interactions between boundaries could be addressed by the development of overarching principles, coordination policies or the influence and steering role of central international organizations, such as the United Nations Environment Programme.[65] Specifically, the author underlines that overarching agreements similar to the Agreement Establishing the World Trade

[56]Commission (EU), 'Europe 2020: A Strategy for Smart, Sustainable and Inclusive Growth' (Communication) COM(2010) 2020 final, 3 March 2010.

[57]Biermann (n 8) 6.

[58]United Nations Framework Convention on Climate Change (adopted 29 May 1992, entered into force 21 March 1994) 1771 UNTS 107.

[59]Convention on Biological Diversity (adopted 5 June 1992, entered into force 29 December 1993) 1760 UNTS 79.

[60]Vienna Convention for the Protection of the Ozone Layer (adopted 22 March 1985, entered into force 22 September 1988) 1513 UNTS 293.

[61]Kim and Bosselmann (n 13) 286–287, 298–302; RE Kim and K Bosselmann, 'Operationalizing Sustainable Development: Ecological Integrity as a Grundnorm of International Law' (2015) 24 Review of European, Comparative and International Environmental Law 194, 200. One of the examples given by these authors of what they call 'environmental problem-shifting' is the replacement of hydrochlorofluorocarbons (HCFCs) with hydrofluorocarbons (HFCs) to reduce the levels of ozone-depleting substances in the atmosphere. Although not harmful to the ozone layer, HFCs have the potential to significantly contribute to global warming.

[62]By 'law', we understand here the set of legal rules issued at the international, regional, national and local levels in every branch of law. In this regard, Ebbesson argues in favour of 'multi-level governance', where 'issues are indeed governed and initiated at different levels and in different ways simultaneously'. J Ebbesson, 'Planetary Boundaries and the Matching of International Treaty Regimes' (2014) 59 Scandinavian Studies in Law 283.

[63]Biermann (n 8) 7.

[64]ibid.

[65]ibid.

Organization, which enshrines overarching principles that regulate different aspects of world trade law,

> *could also be conceived for the governance of the earth system and the protection of the planetary boundaries. However, this would take the form of general statements of principles and the set up of scientific assessment and advisory bodies rather than of a more detailed global framework agreement that covers all boundaries.*[66]

Finally, Biermann proposes to recognize certain standards that are essential for not exceeding planetary boundaries as having the status of *jus cogens*.[67]

Kim and Bosselmann connect the planetary boundaries concept to the notion of ecological integrity and uphold the idea that protecting the integrity of Earth's life-support system should be recognized as an overarching goal and a *grundnorm* of international environmental law.[68] According to them, 'the concept of a *grundnorm* is commonly understood as a fundamental legal principle or norm against which all other legal norms can be assessed and validated. More specifically, it should be seen as a basic norm to bind and guide governmental power.'[69] To facilitate the recognition and implementation of this fundamental norm, these authors propose a major reform in international environmental governance that would be grounded in what they call 'a new constitution-type agreement' or 'an international environmental constitution or covenant'.[70] An international organization would promote its implementation and would have the means to enforce it. Furthermore, States would have the opportunity to prosecute other States before a court of justice, whose judgments would be legally binding. Such action may even involve sanctions.[71] However, given the most recent outcomes of international negotiations on a global legal framework for climate change, namely the 2015 Paris Agreement,[72] it seems to us that the perspective of establishing a punitive legal regime, enforced by an overarching institution, is currently unrealistic. In Paris, parties were very far from reaching a consensus on the establishment of an enforcement mechanism. Instead, Article 15 of the Paris Agreement established a mechanism to facilitate implementation of and promote compliance with the provisions of the Agreement, which shall consist of a committee that shall be expert-based and facilitative in nature and function in a manner that is transparent, non-adversarial and non-punitive.

From our point of view, an effective means for integrating the planetary boundaries approach into international law could be the adoption of a framework convention on planetary boundaries. The 'framework convention' concept has been defined as 'a type of treaty which establishes broader commitments for its parties and leaves the setting of specific targets either to subsequent more detailed agreements (usually called protocols) or to national legislation' and that serves in essence 'as an umbrella document which lays down the principles, objectives and the rules of governance of the treaty regime'.[73] In view of the current scientific uncertainties that flow from the dynamic and complex character of ecosystems and socio-ecological systems at all scales, such a framework convention (that would set the general purpose and principles of a new cooperation and coordination between States and institutions) would be more suitable than any other legal instrument establishing more detailed commitments. Such a legal tool would have the advantage of inherent flexibility,[74] allowing norms to adapt and evolve as necessary in response to the latest scientific advances. In this respect, the 2015 update of the 2009 planetary boundaries framework demonstrates the need for a high adaptive capacity of law.[75]

A framework convention on planetary boundaries would also have the advantage of creating an overarching view of the various existing international agreements and their regimes, including not only those with the explicit purpose of environmental protection but also those that pursue other objectives but that have negative impacts on the environment. A major asset of such a convention would be to set a common and overarching global objective: to maintain the Holocene-like conditions of the Earth system by staying within determined planetary boundaries. Far from creating a new sectoral international legal regime that would complete, overlap, contradict or even duplicate already existing regimes, such a convention would need to develop the framework for identifying the necessary reforms and new standards for each existing legal regime in order to guarantee the coherence of these regimes with the aforementioned common objective.

We argue that a framework convention on planetary boundaries should contain at least the following fundamental elements:

1. The recognition that Holocene-like conditions are the only conditions that we know for certain are favourable for the development of contemporary human societies and their individuals.
2. The recognition that maintaining the Holocene-like Earth system state is a necessary prior condition for the fulfilment of other

[66]ibid.

[67]ibid 8.

[68]Kim and Bosselmann (n 13) 288. See also P Bridgewater, RE Kim and K Bosselmann, 'Ecological Integrity: A Relevant Concept for International Environmental Law in the Anthropocene?' (2015) 25 Yearbook of International Environmental Law 73, 75.

[69]Kim and Bosselmann (n 61) 205. See also Kim and Bosselmann (n 13) 290.

[70]Kim and Bosselmann (n 61) 206.

[71]ibid 206–207.

[72]Paris Agreement (adopted 12 December 2015, entered into force 4 November 2016) 55 ILM 740.

[73]Committee on Housing and Land Management of the United Nations Economic Commission for Europe, 'Framework Convention Concept', seventy-second session, informal notice 5, October 2011. For a definition in French, see AC Kiss, 'Les Traités-Cadre: Une Technique Juridique Caractéristique du Droit International de l'Environnement' (1993) 39 Anuaire français de droit international 793. 'Moving to, or modifying, an umbrella agreement for international environmental law' has been mentioned by Bridgewater and colleagues as a means to improve the visibility and effectiveness of ecological integrity as a legal norm. Bridgewater et al (n 68) 77.

[74]Committee on Housing and Land Management of the United Nations Economic Commission for Europe (n 73).

[75]JB Ruhl, 'General Design Principles for Resilience and Adaptive Capacity in Legal Systems – With Applications to Climate Change Adaptation' (2011) 89 North Carolina Law Review 1373; CA Arnold and LH Gunderson, 'Adaptive Law and Resilience' (2013) 43 Environmental Law Reporter News and Analysis 10426; AS Garmestani et al, 'Can Law Foster Social–Ecological Resilience?' (2013) 18 Ecology and Society 37.

objectives related to life and the well-being of humankind, such as poverty reduction, food security, access to natural resources, economic growth and peacekeeping.

3. The recognition of the existence of ecological thresholds ('tipping points') at the planetary scale and of the fact that humans have become a geophysical force capable of provoking the transgression of such thresholds.
4. The institution of the common global objective of maintaining the biophysical processes that determine the ability of the planetary system to preserve its Holocene-like conditions by ensuring that the cumulative effects of the human-caused impacts on these processes remain at a safe distance, far from any risk of transgressing the planet's ecological thresholds.
5. The recognition of the need to further improve scientific knowledge on the biophysical processes of the Earth system, particularly those with variations that could lead or contribute to the transgression of ecological thresholds, including the identification of such processes, their multi-scale character and their interactions.
6. The recognition that there are multi-scale interactions between the different biophysical processes of the Earth system and, thus, the recognition of the need to manage them in an integrated and coordinated manner that is capable of evolving and adapting according to the most recent scientific progress.
7. The recognition of the need to determine, for each key Earth system process, a scientifically based and quantitative boundary level, or ceiling value, all of which would constitute the planetary boundaries within which human activities can safely take place without jeopardizing the stability of the planetary system.
8. The establishment of an obligation to apply the precautionary principle for determining those boundary levels.
9. The creation of an international panel of experts similar to the Intergovernmental Panel on Climate Change and the Intergovernmental Science–Policy Platform on Biodiversity and Ecosystem Services, which would operate at the interface between science and policy and which would be in charge of the following:

   – Reporting scientific advances in knowledge related to issues such as the functioning of the Earth system, tipping points, the resilience of the planet and the interactions between the various biophysical processes at all scales.
   – Assessing the positive and negative impacts of legal regimes established by the current international and regional agreements on each of the biophysical processes that determine the ability of the Earth system to self-regulate.
   – Providing recommendations for coordinating and reinforcing the coherence between the aforementioned legal regimes to guarantee the achievement of the common global goal established by this framework convention and to ensure that decisions made within the framework of those regimes take into account their eventual impacts on the planet's key biophysical processes.
   – Providing recommendations related to those global biophysical processes that are not yet subject to regulation under international law, such as ocean acidification.

10. The recognition of the ceiling values that constitute the planetary boundaries, together with a 'floor' that represents the respect of human and fundamental rights and incorporates issues related to intra- and intergenerational equity and justice.
11. Finally, the recognition that coordinated efforts must be realized at all scales (global, regional, national and local), taking into account the global character of certain processes (e.g., climate change and ocean acidification) and, on the other hand, the local or regional nature of others (e.g., freshwater use and land system change).

Our proposal for a framework convention on planetary boundaries mirrors certain aspects of the abovementioned proposals of Biermann as well as Kim and Bosselmann in the sense of establishing an overarching goal of Earth system governance and of international law, establishing overarching principles and requiring assessments by international scientific panels. Our proposal differs, however, in terms of the vehicle used for achieving this goal (a framework convention for us) and in the fact that we move the reflection one step forward by developing in greater detail the elements that such a convention should contain.

Our proposal differs in its purpose from existing initiatives that aim to enhance environmental protection through the adoption of international instruments. This is the case, for example, for the proposal for a Global Pact for the Environment, which aims to establish a common, crosscutting and universal core of legally binding general principles of environmental law.[76] Although the preliminary draft of the Global Pact mentions 'the conservation, protection and restoration of the integrity of the Earth's ecosystem' as part of everyone's duty to take care of the environment, and even though the draft also establishes an obligation for parties regarding the resilience of ecosystems and human communities, no tailored guidance is given for effectively safeguarding the Earth system and its key processes in a systemic manner. Another example is the Draft International Covenant on the Human Right to the Environment submitted to the United Nations Human Rights Council at its 34th session by the International Centre of International Comparative Environmental Law.[77] This proposal aims for the adoption of a third international covenant on human rights devoted to the rights of human beings regarding the environment. None of these initiatives is explicitly grounded on and designed for the purpose of maintaining the Holocene-like conditions of the Earth system by staying within determined planetary boundaries, even though they can partially contribute to its achievement.

[76]United Nations Environment Programme (UNEP), 'Concept Note on the Global Pact for the Environment' (11 January 2018) <https://wedocs.unep.org/bitstream/handle/20.500.11822/22458/Global%20Pact%20for%20Environment%20Concept%20Note.pdf?sequence=2&isAllowed=y>; UNEP 'Note on the Global Pact for the Environment' (15 February 2018) <https://wedocs.unep.org/bitstream/handle/20.500.11822/22903/Global%20Pact%20Note.pdf?sequence=4&isAllowed=y>.

[77]HRC 'Written Statement Submitted by the Centre International de Droit Comparé de l'Environnement, a Non-governmental Organization in Special Consultative Status' UN Doc A/HRC/34/NGO/60 (15 February 2017).

4 | CONCLUSION

The inherent complexity of the functioning of the Earth system, including the multi-scale interactions between its different biophysical processes, is reflected in the complex tasks of Earth system governance and of the establishment of a global legal regime committed to protecting and maintaining the system in a favourable state for human development and well-being. In this context, the scientific concept of planetary boundaries has emerged as a useful tool for navigating the complexity of the Earth system, despite all the uncertainties introduced by the present data and knowledge gaps.

Despite the efforts made by the United Nations Secretary-General to promote the planetary boundaries concept among the international community, the concept was included in neither the Rio+20 outcome document[78] nor the 2030 Agenda for Sustainable Development.[79] Only the European Union has successfully introduced the concept in its legal instruments; however, no specific content has been assigned to it, apparently because a stronger scientific basis is desired before moving forward. Nevertheless, humanity cannot afford to wait to have full scientific certainty to address Earth system issues with a systemic approach rather than the ongoing sectoral approach.

We argue that a framework convention could be well suited for integrating the planetary boundaries approach into international law. Because of its generality, such a legal instrument could serve as an ideal means for establishing the common global objective of maintaining the Holocene-like conditions of the Earth system by staying within determined planetary boundaries and for federating all international law under this general purpose and its related principles. Its flexibility could also be useful for progressively adapting norms as new scientific knowledge emerges.

The process recently launched by a United Nations General Assembly resolution for identifying and assessing possible gaps in international environmental law and environment-related instruments, as well as for discussing possible options for addressing these gaps,[80] can be observed as a new opportunity to raise the need for the international community to address Earth-system-related issues in a systemic manner and to introduce and make use of the concept of planetary boundaries for this purpose. Although the General Assembly resolution seems to be focused (as its title suggests) on the objective of adopting an international instrument as the proposed ‘Global Pact for the Environment’, the preliminary draft of which is very distant from an Earth system approach, it is critical to identify and address the gaps related to the effective protection of the entire Earth system and its various key biophysical processes. This could be the first step toward a framework convention on planetary boundaries.

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[78]‘The Future We Want’ (n 37).

[79]UNGA (n 42).

[80]UNGA ‘Towards a Global Pact for the Environment’ UN Doc A/RES/72/277 (10 May 2018). In the follow-up of the proposal for a Global Pact for the Environment, the United Nations General Assembly adopted this resolution, by which it requested the Secretary-General to submit to the General Assembly at its seventy-third session in 2018 a technical and evidence-based report that identifies and assesses possible gaps in international law and environment-related instruments with the aim of strengthening their implementation, and decided to establish an ad hoc open-ended working group under the auspices of the General Assembly to consider the report and discuss possible options to address possible gaps identified in it and, if considered necessary, discuss the scope, parameters and feasibility of an international instrument with the aim of making recommendations, which may include the convening of an intergovernmental conference to adopt an international instrument, to the Assembly during the first half of 2019.